

MORE COMPLICATED EQUATIONS

More complicated equations are rearranged in the same way as simple equations—anything done to simplify one side of the equation must also be done to the other side so that both sides of the equation remain equal. The equation will give the same answer no matter where the rearranging is started.

Example 1

This equation has numerical and unknown terms on both sides, so it needs several rearrangements to solve.

First, rearrange the numerical terms. To remove the -9 from the right-hand side, add 9 to both sides of the equation.

Next, rearrange so that the a 's are on the opposite side to the number. This is done by subtracting $2a$ from both sides.

Then rearrange again to make a the only subject of the equation. Since the equation contains $3a$, divide the whole equation by 3.

The subject of the equation, a , is now on its own on the right side of the equation, and there is only a number on the other side.

Reverse the equation to show the unknown variable (a) first. This does not affect the meaning of the equation, because both sides are equal.

numerical term a appears on both sides of the equation there are numerical terms on both sides

$$3 + 2a = 5a - 9$$

add 9 to 3 add 9 to -9 , which leaves 0 so $5a$ is isolated

$$12 + 2a = 5a$$

$2a - 2a = 0$, leaving 12 on its own $5a - 2a = 3a$

$$12 = 3a$$

the right side must be divided by 3 to isolate a , so the left side must also be divided by 3 to keep both sides equal divide $3a$ by 3 to leave a on its own

$$\frac{12}{3} = \frac{3a}{3}$$

$12 \div 3 = 4$, which is the value of a a is now the subject, isolated by itself on one side of the equation

$$4 = a$$

put the variable first this is the solution of the equation—it gives the value of the variable (a)

$$a = 4$$

Example 2

This equation has unknown and numerical terms on both sides, so it will take several rearrangements to solve.

First rearrange the numerical terms. Subtract 4 from both sides of the equation so that there are numbers on only one side.

Then rearrange the equation so that the unknown variable is on the opposite side to the number, by adding $2a$ to both sides.

Finally, divide each side by 8 to make a the subject of the equation, and to find the solution of the equation.

there are numerical values on both sides there are terms including the unknown a on both sides

$$6a + 4 = 5 - 2a$$

$4 - 4 = 0$, so $6a$ is on its own take 4 from 5, leaving 1

$$6a = 1 - 2a$$

$6a + 2a = 8a$ $-2a + 2a = 0$, so 1 is on its own

$$8a = 1$$

divide $8a$ by 8 to leave a by itself on the left of the equation because the left side was divided by 8 to isolate a , the right side must also be divided by 8 to keep both sides equal

$$a = \frac{1}{8}$$